

Conflict and Peace

Week 14: Personnel capacity of the state

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Why does the public sector personnel matter?

Take up huge roles: In principle and in data

- As in other sectors, they embody human capital of the state
- They determine provision of public services and implementation of policies
- Takes up nearly 20% of all employment and 35% of formal employment (World Bank)
- Responsible for the public sector that takes up approximately 45% of GDP (OECD)

However, the quality of public sector workers varies across countries

- 20-35% of front-line workers (teachers, health care) absent in developing countries
- Lack institutional structures to incentivize, monitor existing workers
 - ▶ Poor structure → Absenteeism and inefficiencies → Bad public services (repeat)
- Finding motivated and capable individuals to work is difficult
 - ▶ Lower pay, more rigid mobility than private sector
- Bureaucratic quality is also clustered with development

Public sector takes up huge share of employment globally

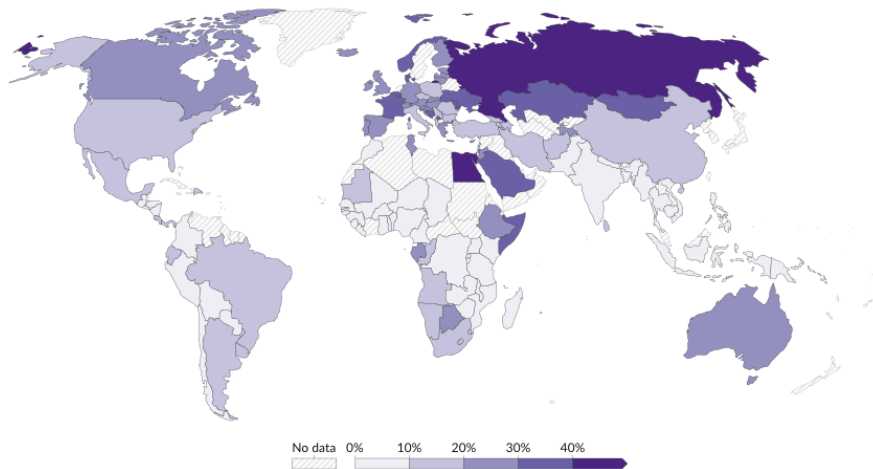
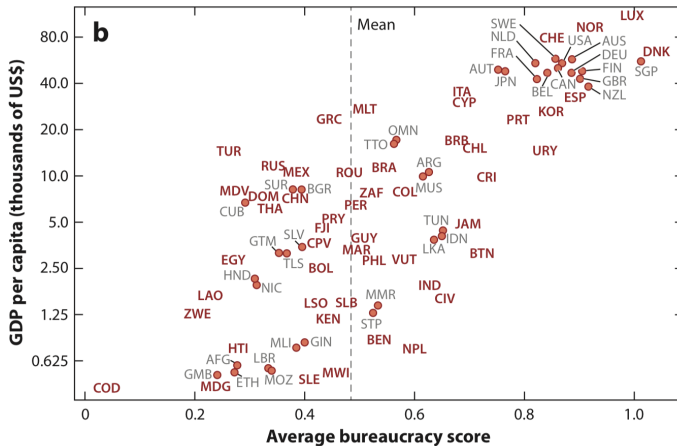


Figure: Share of public sector employment in total employment, 2019

Source: Our World in Data ([link](#)), reprocessed from World Bank and ILO data

Clustering of development and bureaucratic quality



Source: Besley, Burgess, Khan, Xu (2022) "Bureaucracy and Development", *Annual Review of Economics*, 14, 397-424

Bureaucrat recruitment and principal-agent problem

Ideal bureaucrats vs reality

- High ability individuals with prosocial motives
- But lower pay → smaller applicant pool, less qualified individuals
- Mixed evidence from field experiments
 - ▶ Motivated applicants (Dal Bó et al 2013) vs Applying 'for money' (Desserano 2015)

Problem with incentivizing, monitoring existing public sector workers

- Public workers as agents, superiors (and citizens) as principal
- In many cases, agents' interest not aligned with principals' interests
 - ▶ Actions of agents are not always observable → pursue own interest
- What structure aligns interests while motivating agents is a big question

We look at various exploring questions on selection, retention, and motivation of personnel in public sector

Roadmap for today

1. Overview of the topic

2. Impact of leadership and politician capacities

- 2.1 Dal Bó, Finan, Folke, Persson, Rickne (2017): Who Becomes a Politician?
- 2.2 Xu (2018): The Costs of Patronage: Evidence from the British Empire
- 2.3 Fenizia (2022): Managers and Productivity in the Public Sector
- 2.4 Jones, Olken (2005): Do Leaders Matter? National Leadership and Growth since World War II [student presentation]

3. Recruiting, incentivizing, and monitoring bureaucrats

- 3.1 Dal Bó, Finan, Rossi (2013): Strengthening State Capabilities: The Role of Financial Incentives in the Call to Public Service
- 3.2 Bandiera, Best, Khan, Prat (2021): The Allocation of Authority in Organizations: A Field Experiment with Bureaucrats
- 3.3 Callen, Gulzar, Hasanain, Khan, Rezaee (2023): The Political Economy of Public Sector Absence [student presentation]

Dal Bó, Finan, Folke, Persson, Rickne (2017):
Who Becomes a Politician?

Q: How to attract competent leaders while being representative?

Can democracies form competent and representative leadership (inclusive meritocracy)?

- Voters want to elect leaders capable of efficiently implementing policies
- Voters also want policy-makers who represent diverse interests
- Classical economic models suggest that these two are in a trade-off relation
 - ▶ Market wage < Returns from office for low-quality individuals (low opportunity cost)
 - ▶ Good politicians likely to 'free-ride' on bad politicians → Supply of good politicians ↓
 - ▶ Even if good candidates can be selected, may make broad representation difficult

This paper uses data on Swedish local politicians to identify patterns in inclusive democracy

- Includes elected and *nonelected*: Address selection bias
- Detailed measures on quality and representation
- Identifies how both competent and representative leaders are selected in democracies

Context and data

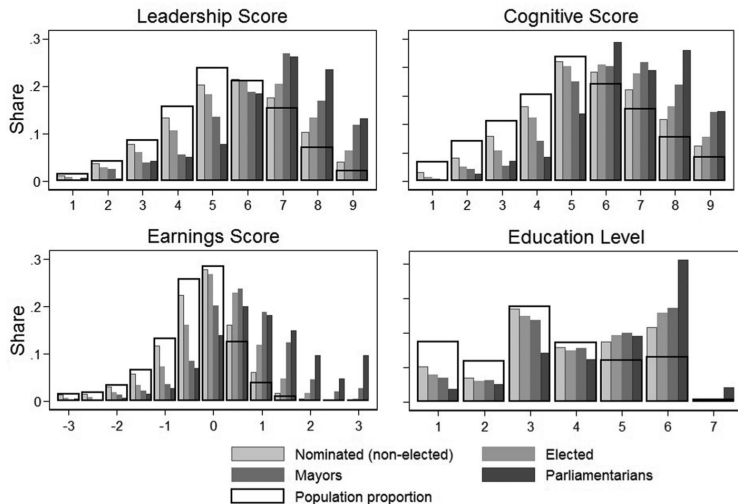
Swedish National MPs and Municipal council members

- MPs are full-time, while most municipal council members have other work
 - ▶ Mayors, selected among council members, are full-time and paid highly
- Low monetary cost to individuals running for local office
- Key motives: Intrinsic concerns with policy, social interaction in policy circles
- Left represents blue-collar, center-right represents white-collar workers

Data obtained from

- Sample: All candidates from 1980-2010
- Links candidates to admin registers (including military enlistment and tax records)
- Age, sex, education level, earnings, occupation, parental/sibling info, cognitive capacity

Positive selection of politicians based on abilities



- Beats classical prediction that there are more low-quality individuals entering politics

Are they socially representative?

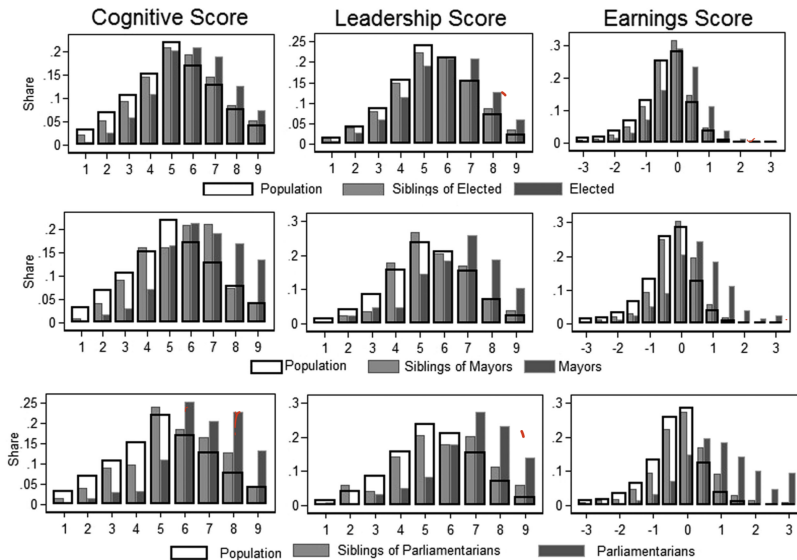
Determining source of positive selection

- Privileged access? Ability should matter little once elite status is controlled for
- Exclusive meritocracy? Key abilities are strongly associated with social background
- Inclusive meritocracy? High ability individuals come from all social backgrounds

Empirical tests

- Compare with siblings: Same "elitist background" → Positive selection still appears
- Parental income: Social background of politicians is evenly distributed
- Social class: Politicians from wide range of occupations represented

Elitism unlikely to explain positive selection



Neither does exclusive meritocracy

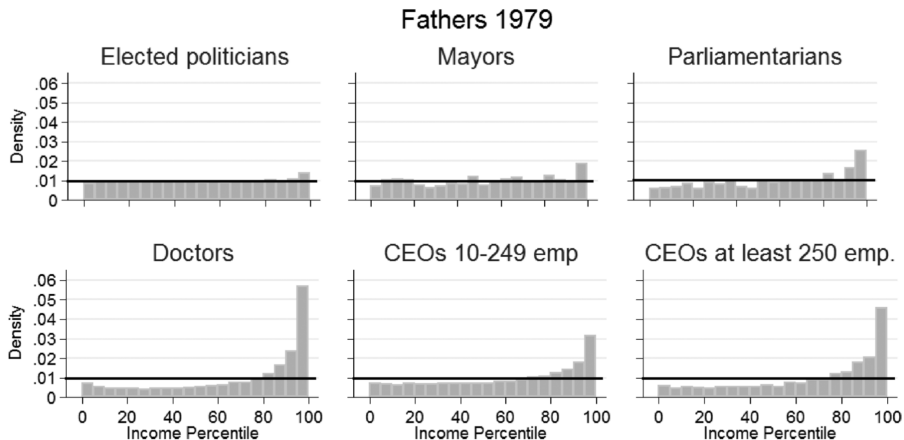


FIGURE IV

Distribution of Fathers of Politicians and Other Occupations across the Percentiles of Population Income

Discussion and takeaways

What drives competence and representation?

- Weak trade-off between competence and social background
- Positive selection on ability even among those with poorer backgrounds
- Intrinsic motives: Positive selection even for low-paid municipal councils
- Screening by parties: Sorts more able people to higher positions

Remaining questions

- When and where do these findings hold outside 1980-2010 Sweden?
 - ▶ Intrinsically motivated individuals may not enter politics elsewhere, how to attract them?
 - ▶ Party incentives: Promote someone who can win vs. who has best policy
 - ▶ Other democratic countries? Sweden now?

Xu (2018):
The Costs of Patronage: Evidence from the
British Empire

Q: How does patronage affect allocation of positions & performance?

How personnel are promoted and incentivized determines organizational performance

- Past: patronage widely used to appoint public sector positions
 - ▶ Discretionary appointments based on network and ties to the ruler
- In theory, discretion can align incentives between principal and agents
- However, it can worsen performance if favoritism disincentivizes capable subordinates

This paper uses historical data to identify how patronage affects performance

- Colonial governors of British empire: 1854-1966 across 70 colonies
- Connection determined by genealogy and biological data
- Within-governor variation: Change in secretary of state → Change in connections
- Across-time variation: Discretion (1854-1930) vs. Independent appointment (1930 -)
- Test who gets assigned to richer colonies and how tax returns are affected

Organization of British colonies

Assignment of colonial governors

- Handled by Colonial Office under Secretary of State for the Colonies in London
- Secretary of State in charge for around 3 years
- 1930 Reform replace Secretary of State's discretion with open recruitment

Role of the governors

- Fixed 6-year term
- Responsible for public finance, legislation, and personnel appointments
- Discretion over executive and finances: Widely different policies across colonies
- Direct monitoring impossible due to distance

Data and empirical strategy

Digitalized (by Guo Xu himself!) archival data

- Sources: Colonial Blue Books, Colonial Office List, Who's Who, The Peerage
- Postings (and their wages), background of governors, genealogy
 - ▶ Connection determined by ancestry, membership at aristocratic clubs, and schools
- Colonial statistics (revenue, expenditure, demographics etc)

Empirical strategy leverages differences in connectedness across different periods

$$y_{it} = \beta c_{it} + \theta_i + \gamma X_{it} + \tau_t + e_{it}$$

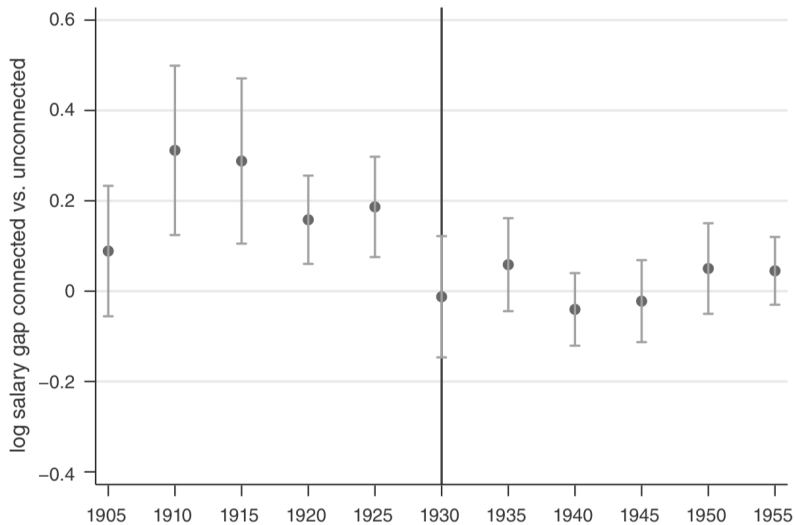
$$y_{it} = \beta_0 c_{it} + \beta_1 c_{it} \times I[t \geq 1930] + \theta_i + \gamma X_{it} + \tau_t + e_{it}$$

- Connection c_{it} used with various criterion
- c_{it} is predetermined and driven by secretary of state changes
 - ▶ No reverse causality: Unlikely to attend Oxbridge to be connected to current governor
 - ▶ Secretary of state turnover generates common shocks to all serving governors

Connected governor salaries are higher!

	log governor salary in GBP					
	(1)	(2)	(3)	(4)	(5)	(6)
Mean of dep. var.	7.929	7.929	7.929	7.929	7.929	7.929
No. colonies served	0.221 (0.035)	0.222 (0.035)	0.223 (0.035)	0.222 (0.035)	0.224 (0.035)	0.223 (0.035)
Shared ancestors	0.103 (0.047)				0.093 (0.046)	
Both aristocrats		0.215 (0.124)			0.176 (0.121)	
Both Eton			0.133 (0.077)		0.118 (0.081)	
Both Oxbridge				0.072 (0.047)	0.073 (0.045)	
Connected						0.097 (0.036)
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Governor FEs	Yes	Yes	Yes	Yes	Yes	Yes
Spell length FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,510	3,510	3,510	3,510	3,510	3,510

Salary gap washed away post-1930



Patronage worsens performance, which is offset following reforms

	Colony-level				Public expenditure			
	Public review		Public expenditures		Overall		Tax	Works
	Overall		Trade	Internal	(5)	(6)	(7)	(8)
	(1)	(2)	(3)	(4)				
Panel A. Revenue					12.33	12.37	9.015	10.32
Mean of dep. var.	12.31	12.31	11.47	11.59	−0.029	−0.042	−0.089	−0.107
Connected	−0.040 (0.017)	−0.055 (0.021)	−0.053 (0.026)	−0.043 (0.032)	(0.019)	(0.023)	(0.053)	(0.062)
Connected × reform dummy		0.061 (0.033)				0.053 (0.034)		
Connected + connected × reform dummy	−	0.005 (0.026)	−	−	−	0.010 (0.025)	−	−
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Governor-Colony FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Spell length FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time-varying controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,510	3,510	2,670	2,657	3,510	3,510	1,742	2,588

Takeaways and discussions

What the findings say about organizational performance

- Connection → Positions that yield higher wages
- Lower revenue collection: Due to higher tax exemptions, high reliance on customs
- Reforms washed salary gaps and shortfalls in performance

Discussions

- Favoritism can lead to worse organizational performance, even in public institutions
- But how does it affect recruitment into the pool of governor candidates?
 - ▶ If people know connections matter, talented individuals with no connection discouraged
 - ▶ Leaving pool of candidates with less talented individuals
 - ▶ Ways to look for 'outsiders' with talent?

Fenizia (2022): Managers and Productivity in the Public Sector

Q: Do talented managers in public sector improve performance?

How important are managers in public sectors?

- Oversee day-to-day operations and supervise policy implementation
- Limited tools available compared to private sector managers"
 - ▶ Limited discretion in firing and promoting workers
- How to measure performance of government agencies?

This paper uses innovative measures of performance to study the role of managers

- Italian Social Security Agency data (INPS): Covering various welfare programs
- Uses welfare claims as measure of output (homogenous across different offices)
- All sites have same rules and physical capital → distinguish manager contributions

Background and data

Role of INPS

- Administers all social welfare and insurance programs in Italy
- Claims are processed in local offices run by single managers
- Managers allocate work, determine workplace practices, but cannot fire workers
- Managers selected among limited pool of candidates and rotate every 5 years

Data sources on managers (2011Q1 - 2017Q2)

- Output: $\sum$ claims weighed by complexity (processing time) per worker
- Service quality: Timeliness and error rate
- Worker characteristics: Job title, age, office locations etc

Empirical strategies

Do productive managers improve performance?

- Fixed effects regression with office(i), time(t), and manager($m(i, t)$) fixed effects

$$\log P_{it} = \alpha_i + \tau_t + \theta_{m(i,t)} + u_{it}$$

- Requires that manager moves are exogenous conditional on office and time effects
 - ▶ Observable characteristics should not predict manager quality
 - ▶ Violated if better managers are assigned to worse offices
- Compare adjusted R^2 to test how much of output variation is explained by managers

What makes for a productive manager?

- Regress changes in outcomes with quarter before manager change to talent changes

$$y_{i,t+k} - y_{i,t-1} = \pi_0^k + \pi_1^k(\hat{\theta}_{i,new}^k - \hat{\theta}_{i,old}) + \gamma^k X_i + (e_{i,t+k} - e_{i,t-1})$$

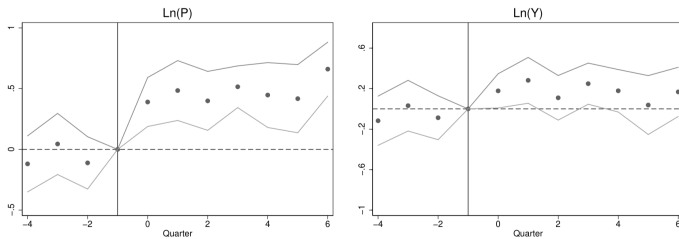
Managers explain 10% of variation in productivity

TABLE IV
ANALYSIS OF VARIANCE OF OFFICE PRODUCTIVITY

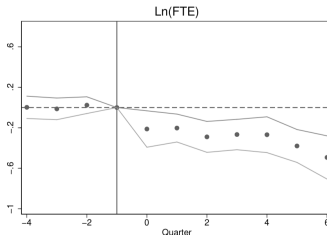
	(1) Ln(P)	(2) Ln(P)	(3) Ln(P)	(4) Ln(P)	(5) Ln(P)
N	3,316	3,316	3,316	3,316	3,316
R sq.	0.325	0.727	0.835	0.789	0.839
Adj. R sq.	0.324	0.679	0.762	0.720	0.765
Time FE	Yes	Yes	Yes	Yes	Yes
Office FE	No	Yes	Yes	No	No
Manager FE	No	No	Yes	Yes	No
Manag-by-Office FE	No	No	No	No	Yes
Pvalue			0.000	0.000	

- Adjusted R^2 : 0.68 in (2) $\rightarrow$ 0.76 in (3): 8 p.p increase
- $0.08/0.762 \simeq 0.104$: Manager explains 10% of variation in productivity
- Robust to selective sorting, other ways of variance decomposition

Talented managers improve productivity, with workers decreasing...



(c)



Talented managers induced by retirement of older workers

k	(1) Retirement	(2) Separations	(3) Hires	(4) Fires	(5) Inbound T	(6) Outbound T
-4	0.041 (0.125)	0.099 (0.111)	0.178 (0.110)	0.003 (0.004)	0.120 (0.121)	-0.039 (0.146)
-3	-0.044 (0.088)	-0.011 (0.078)	0.093 (0.090)	0.002 (0.004)	0.069 (0.090)	-0.111 (0.133)
-2	-0.055 (0.059)	0.033 (0.053)	0.034 (0.073)	0.003 (0.004)	0.068 (0.082)	-0.122 (0.112)
0	0.301 (0.085)	0.038 (0.081)	0.024 (0.018)	-0.008 (0.010)	0.031 (0.159)	-0.023 (0.053)
1	0.393 (0.100)	-0.011 (0.099)	0.027 (0.033)	-0.056 (0.031)	-0.033 (0.163)	-0.010 (0.067)
2	0.381 (0.105)	-0.057 (0.117)	0.024 (0.033)	-0.049 (0.040)	-0.196 (0.172)	-0.142 (0.097)
3	0.392 (0.117)	-0.012 (0.126)	0.006 (0.038)	-0.063 (0.045)	-0.327 (0.174)	-0.234 (0.114)
4	0.438 (0.116)	-0.0722 (0.128)	-0.015 (0.039)	-0.040 (0.038)	-0.385 (0.171)	-0.231 (0.123)
5	0.413 (0.120)	0.015 (0.135)	0.005 (0.039)	-0.061 (0.041)	-0.403 (0.174)	-0.303 (0.125)
6	0.399 (0.125)	-0.121 (0.155)	-0.082 (0.057)	-0.059 (0.042)	-0.537 (0.184)	-0.405 (0.135)
N	318	318	318	318	318	318
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

Takeaways and discussions

Other outcomes

- Trade-off in quality? Not observed
- Outcomes not driven by shifting towards easier claims to handle
- Thus managers do have a meaningful impact on productivity

Remaining questions

- The managers oversee relatively small teams with well-defined goals this context?
 - ▶ What about large projects and teams?
 - ▶ What about organization with no fixed roles and less structure?
- Other factors may determine managers' role in productivity
 - ▶ Ability to obtain inputs efficiently (no bribing, better communication)
 - ▶ Ability to attract more talented workers under the command

Jones, Olken (2005):
Do Leaders Matter? National Leadership and
Growth since World War II [student
presentation]

Dal Bó, Finan, Rossi (2013):
Strengthening State Capabilities: The Role of
Financial Incentives in the Call to Public Service

Q: Can fiscal incentives attract capable individuals to public sector?

Professionalized bureaucracy require individuals with capabilities and integrity

- These do not always co-move
 - ▶ Capable individuals may seek opportunities in private sector (higher pay)
 - ▶ This may leave public sector with less able individuals
 - ▶ Not all capable individuals have integrity:
- So what makes up motivated agents and how to attract them to public sector?

This paper investigates role of incentives while investigating various aspects of quality

- How do monetary and non-monetary incentives attract candidates?
- Estimates if higher wages attract more candidates and those of higher quality
- Also estimates the labor supply elasticity facing public sector in Mexico
- Heterogeneity by locational characteristics (local crime rates)

Context of the experiment

Mexico's regional development program (RDP) in 2011

- Network of 350 agents & 50 coordinators to identify needs, report to federal govt
- Exogenously assign different wage offers across different posts (5000 vs 3750 MXP)
- Candidates screened to document various characteristics
 - ▶ Aptitude: intelligence (IQ) and other qualities known to determine market value
 - ▶ Motivation: Integrity, prosocial inclinations, public service motivation (interviews)
 - ▶ Attraction & commitment to policy making, justice, civicness, compassion, self-sacrifice
- Vacancies randomized to one of 4 strata → Offers randomized to indiv. in that stratum
 - ▶ High vs low wage, High (> 9) vs normal IQ (7-9 Ravens matrix)

Estimating causal effects: OLS (due to random assignment design)

- Effect on character Y in applicant pool (region): $Y_{icr} = \beta_1 T_c + \xi_r + e_{icr}$
- Effect on acceptance A among applicants (strata): $A_{ics} = \gamma_1 T_c + \beta X_i + \xi_s + e_{ics}$
- Heterogeneity by municipalities: $A_{ics} = \gamma_1 T_c + \gamma_2 T_c \times W_m + \gamma_3 W_m + \beta X_i + \xi_s + e_{ics}$

Higer wage → High ability applicants ↑

	Observations (1)	Control (2)	Treatment effect (3)
Number of applicants	106	18.093	4.714 [4.430]
Panel A: Market skills			
Wage in previous job	1,572	3479.667	819.154 [174.703]***
Previous job was white collar	1,170	0.243	0.069 [0.029]***
Currently employed	2,225	0.104	0.053 [0.019]***
Has work experience	2,212	0.459	0.167 [0.048]***
Years of experience in past 3 spells	2,212	1.185	0.284 [0.171]
IQ (Raven test)	2,229	8.488	0.506 [0.223]**
Raven score ≥ 9	2,229	0.572	0.091 [0.039]**
Chose dominated risk option	2,213	0.431	-0.064 [0.025]**
Years of schooling	2,198	14.552	0.091 [0.308]

	Observations (1)	Control (2)	Treatment effect (3)
Panel B: Personality traits			
Extraversion	2,189	3.674	0.013 [0.036]
Agreeableness	2,167	4.107	0.004 [0.022]
Conscientiousness	2,191	4.235	0.063 [0.030]**
Neuroticism	2,168	2.254	-0.099 [0.033]***
Openness	2,168	3.910	0.042 [0.028]
Big 5 index	2,099	0.000	0.087 [0.049]*
Integrity: direct	2,223	0.067	-0.009 [0.013]
Integrity: indirect	2,099	44.424	0.602 [1.232]

Higer wage → Better motivated applicants ↑↑

				Panel B: Prosocial behavior		
				Altruism	2,199	23.491
						0.039
						[0.291]
				Negative reciprocity	2,206	0.508
						0.075
						[0.023]***
				Cooperation	2,157	26.174
						0.675
						[0.404]*
				Did charity work	2,223	0.605
				in the past year		−0.096
						[0.041]**
				Volunteered in	2,224	0.710
				the past year		−0.006
						[0.027]
				Importance of wealth	2,025	3.159
						0.107
						[0.087]
				Belongs to a political party	2,225	0.113
						−0.026
						[0.014]*
				Voted	2,225	0.758
						0.019
						[0.035]

Higer wage → Applicants more likely to accept

	Accepted (1)	Accepted (2)	Rejected (3)	Not reachable (4)
High-wage offer	0.151 [0.054]***	0.160 [0.054]**	-0.017 [0.034]	-0.135 [0.054]***
Characteristics				
Male		-0.080 [0.058]		
Years of schooling		-0.022 [0.008]**		
High IQ		-0.017 [0.053]		
Wage in previous job > 5,000 pesos		-0.007 [0.067]		
Big 5 index		0.042 [0.038]		
PSM index		-0.024 [0.044]		
Mean of dependent variable	0.55	0.55	0.13	0.32
Observations	350	343	350	350
R-squared	0.10	0.12	0.09	0.13

Higher wage needed to offset disadvantageous regional traits

Dependent variable	Acceptance				
	(1)	(2)	(3)	(4)	(5)
High-wage offer	0.047 [0.037]	0.066 [0.047]	0.075 [0.053]	0.056 [0.041]	0.147 [0.047]***
High wage offer $\times$ Distance	0.026 [0.005]***			0.028 [0.007]***	0.010 [0.007]
Distance	-0.027 [0.004]***			-0.028 [0.007]***	-0.017 [0.005]***
High wage offer $\times$ Drug-related deaths per 1,000 inhabitants		0.078 [0.039]*		-0.033 [0.037]	-0.025 [0.040]
Drug-related deaths per 1,000 inhabitants		-0.107 [0.044]**		0.000 [0.045]	-0.022 [0.040]
High wage offer $\times$ Human Development Index			-1.482 [0.738]**	-0.913 [0.645]	-1.368 [0.589]**
Human Development Index			1.526 [0.673]**	1.044 [0.598]*	1.002 [0.521]*
Observations	238	238	238	238	348
R-squared	0.25	0.21	0.2	0.26	0.18

Additional findings and takeaways

Labor supply elasticity (ϵ) for public sector with respect to wages

- $\epsilon = \epsilon_{\text{pool}} + \epsilon_{\text{accept}} + \epsilon_{\text{pool}} \times \epsilon_{\text{accept}} \times \text{wage difference}$
- Wage difference: 33% $(=(5000-3750)/3750)$
- Pool: 4.7 person $\uparrow$ from 18.1 person (26.3% $\uparrow$) (Table 3) $\rightarrow$ Elasticity: 0.8 $(=26.3/33)$
- Acceptance: 15.1 p.p $\uparrow$ from 42.9% (35.2% $\uparrow$) (Table 5) $\rightarrow$ Elasticity: 1.07 $(=35.2/33)$
- $0.8 + 1.07 + 0.8 \times 1.07 \times 0.33 = 2.15$ (1% $\uparrow$ in wages $\rightarrow$ Supply 2.15% $\uparrow$)

Takeaways

- Offering high wages attracts better and motivated candidates
- However, findings hold only if motivation and ability are positively correlated
 - ▶ What if service commitment is sacrificed for higher pay (prioritize superiors over citizens?)
- Silent on post-acceptance performance: High pay to keep people motivated once hired?

Bandiera, Best, Khan, Prat (2021):
The Allocation of Authority in Organizations: A
Field Experiment with Bureaucrats

Q: How does allocation of authority to bureaucrats affect outcomes?

Organization bring together people with different interests and skills

- Working towards a common goal challenging: Principal-agent problem and monitoring
- How to allocate decision rights at different layers of hierarchy?
- How to monitor and incentivize agents' behavior?
- Many work on effects of performance pay, few look at decision-making structure

This paper designs an experiment in Pakistan procurement sector that combines the two

- Strict rules and monitoring vs. simplification and more autonomy?
- Procurement officers (PO, make purchases) and Monitors (approve purchase)
 - ▶ Monitors: Approve purchases quickly vs Delay purchase until end of fiscal year
- Experiment: Control + Autonomy + Performance pay + Both
- Different policy instrument works in different structures

Procurement structure in Pakistan and theoretical framework

Procurement structure

- Experiment took place in 26 / 36 districts in Punjab (80% of population)
- Each office has 1 PO with allocated budgets
- POs submit purchase for pre-audit approval to a monitor in each district
 - ▶ This monitor can ask for extra paperwork (withhold) or approve the purchase

Conceptual framework: Expected effects of autonomy vs incentives by type of monitors

- Aligned PO and monitors behave in the interest of citizens (low price, little delays)
- Misaligned PO raises price (bribes, little effort to find low-price)
- Misaligned monitor delays purchase (exert power on PO)
- Autonomy: Price↓ if PO more aligned then monitor (opposite if PO more misaligned)
- Incentives: Motivate PO, but dampened with misaligned monitors (higher costs)

Experimental design

Measuring bureaucratic performance with value for money (with online platform)

- Price conditional on quantity, nature of purchased goods, and net of transaction costs
 - ▶ Items: Consumables such as papers, toners, pen, banners, light bulbs etc
 - ▶ Price is adjusted for delivery speed, cost of delivery
- Others: Expected price, good-bad purchase categorization(based on attributes), ML

Experimental design

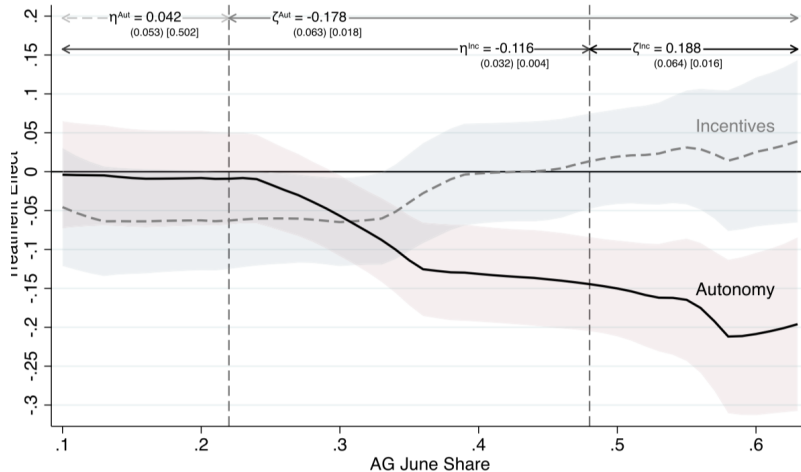
- Autonomy: Remove the role of monitors in purchasing decisions
- Incentive: Range of bonus for improving value for money ($\times 2$ - $\times 1/2$ of monthly wage)
- Another group that receives both
- Stratified by district-department level

$$y_{igto} = \alpha + \sum_{k=1}^3 \eta_k \text{Treatment}_o^k + \rho_g q_{igto} + \delta_s + \gamma_g + e_{igto}$$

Overall, autonomy → lower prices paid but not incentives

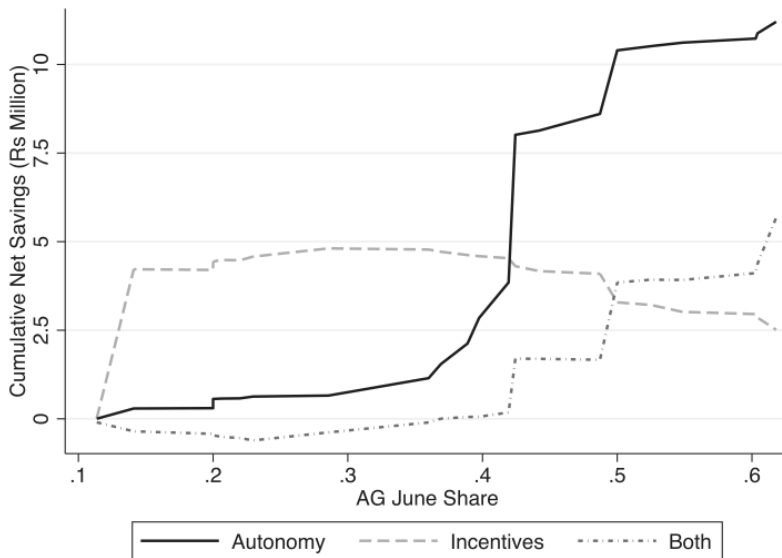
	Variety			Unit price				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Autonomy	0.016 (0.030) [.646]	0.010 (0.023) [.705]	-0.000 (0.009) [.819]	-0.085 (0.038) [.046]	-0.086 (0.032) [.018]	-0.080 (0.031) [.023]	-0.082 (0.034) [.030]	-0.085 (0.038) [.047]
Incentives	0.006 (0.030) [.846]	0.025 (0.023) [.325]	0.005 (0.009) [.382]	-0.016 (0.038) [.723]	-0.026 (0.030) [.476]	-0.022 (0.033) [.571]	-0.020 (0.034) [.625]	-0.017 (0.038) [.717]
Both	0.037 (0.030) [.265]	0.059 (0.023) [.021]	0.006 (0.009) [.484]	-0.070 (0.041) [.130]	-0.083 (0.032) [.025]	-0.072 (0.033) [.053]	-0.086 (0.039) [.043]	-0.070 (0.041) [.127]
Item variety measure	Scalar	Coarse	ML	None	Attribs	Scalar	Coarse	ML
$p(\text{all} = 0)$	.660	.080	.777	.168	.054	.093	.087	.165
$p(\text{autonomy} = \text{incentives})$	.749	.537	.447	.146	.077	.119	.119	.152
$p(\text{autonomy} = \text{both})$	.461	.031	.566	.741	.927	.807	.932	.747
$p(\text{incentives} = \text{both})$	.302	.144	.953	.262	.133	.227	.136	.257
Observations	11,771	11,771	11,771	11,771	11,771	11,771	11,771	11,771

Effects differ by type of monitors



- Better monitors (low delays, AG June share ↓): Incentive lower price (not autonomy)
- Worse monitors (more delays, AG June share ↑): Autonomy lower price (not incentives)

Treatment → large savings, but heterogenous by monitor types



Takeaways and remaining questions

Incentives and autonomy work in different situations

- Incentives to low-level agents mean little if their supervisors are misaligned
- In such case, giving low-level agents more autonomy improves performance
- If supervisors are aligned, incentives improve effort of agents
- On the other hand, autonomy does not change much from status quo in this case

Remaining questions on the organizational Economics

- Can stringent rules, not autonomy, be an optimal choice?
 - ▶ Could still be worthwhile to prevent misaligned agents imposing high costs
- How does autonomy affect *who becomes a bureaucrat*?
 - ▶ How to bring in more aligned agents (make efficient decisions for public) than misaligned agents (extract personal benefits with autonomy)?

Callen, Gulzar, Hasanain, Khan, Rezaee (2023):
The Political Economy of Public Sector Absence
[student presentation]

Takeaways and remaining questions

Takeaways

- Better leaders and bureaucrats improve performance of organizations
- Personnel selected on factors other than merit can hurt performance
- Different organizational structure works best to align principal-agent interests
 - ▶ If lower-ranked agents are more aligned than middle-managers, more autonomy helps
 - ▶ If different levels of bureaucracy are similarly aligned, incentives induces better work
- Recruiting capable & motivated agents, while monitoring & incentivizing workers are difficult to balance

Remaining questions

- Delineation of roles left to 'politicians' and 'bureaucrats'
 - ▶ Perform different roles and subject to different accountability measures
 - ▶ What roles are better left to politicians vs bureaucrats?
- Relationship with private sector
 - ▶ Complement developmental efforts vs limit/capture each other?